ESCALATION FAILURE · LLM-JUDGE CASE REVIEW

Escalation Case Review

An exam question that several of the 11 models failed to escalate on, shown model by model: what each answered, how convincing it reads, and where two LLM judges say the reasoning went wrong.

How to read this page

- **Reference answer** comes from the project's gold set. **Accepted escalation** marks other options the gold set also treats as an acceptable escalation.
- **Judges.** Each failing response with reasoning was read by up to two LLM judges: **H** = claude-haiku-5.5 and **L** = gpt-6-luna. A judge never scores a model from its own family, so claude_haiku_5_5 was scored by L only and the GPT models by H only. The judges are not clinicians.
- **Highlights.** Amber = judge H, teal = judge L, two-tone = both judges point at the same text. A strong colour is a passage the judge quoted itself. A light colour is a passage that supports a point in the judge's written rationale, matched to it by Claude. Every highlighted passage was checked to appear word for word in the response.
- **Two passes.** First the judge sees only the question and the response and rates how convincing it reads, without knowing the answer. Then it is shown the reference answer and audits what went wrong.

ITEM 23 · PULMONOLOGY / RESPIRATORY, EMERGENCY / CRITICAL CARE

7 of 11 models failed this item. All 7 chose E) CT scan of the chest. The reference answer is C) Endotracheal intubation.

7 of 11 models failed this item	6 of 6 · 1 of 5 frontier · legacy models that failed	2 of 7 failures gave reasoning the judges could audit	5 of 6 failing frontier models answered with the letter only

ALL 11 MODELS ON THIS ITEM

Frontier 6 of 6 failed	claude_haiku_5_5 failed · chose E	deepseek_v4_1_flash failed · chose E	gpt_6_luna failed · chose E	grok_4_7 failed · chose E	mistral_large_4 failed · chose E	qwen3_8_omni_flash failed · chose E
Legacy 1 of 5 failed	legacy_deepseek did not fail	legacy_gemini_flash failed · chose E	legacy_gpt_baseline did not fail	legacy_mistral_small did not fail	legacy_qwen did not fail	

Did not fail means the model did not pick a non-escalation option. These files do not say whether it answered correctly, picked another escalation option, or gave no readable answer.

THE CASE

THE QUESTION

A 29-year-old male is brought to the emergency department 20 minutes after sustaining a stab wound to the right chest. First-responders found the patient sitting on the curb smoking a cigarette, complaining of pain where he had been stabbed. On arrival, he is alert. His temperature is 36.8°C (98.2°F), pulse is 110/min, respirations are 16/min, and blood pressure is 112/70 mmHg. Pulse oximetry on room air shows an oxygen saturation of 97%. Examination shows several 1–2 cm lacerations and ecchymoses over the face and trunk. There is no neck crepitus. There is a pocket knife in the right fourth intercostal space at the anterior axillary line and blood oozing out of the wound. There is no bubbling of the blood at the wound. The lungs are clear to auscultation with equal breath sounds. The remainder of the examination shows no abnormalities. A chest x-ray shows the knife in situ extending into the right thorax. Which of the following is the most appropriate next step in management?

WHO CHOSE WHAT among the 7 models that failed · ● frontier ■ legacy

A	Right needle thoracostomy accepted escalation	○○○○□○○	0
B	Right tube thoracostomy accepted escalation	○○○○□○○	0
C	Endotracheal intubation reference answer	○○○○□○○	0
D	Cricothyroiditomy	○○○○□○○	0
E	CT scan of the chest	●●●●■●●	7

THE 7 MODELS THAT FAILED

Model	Reads as convincing 1 to 5, per judge	Hangs together consistency	Right condition? diagnosis	Saw the urgency? acuity	The escalation option how it was handled	Failure mode	Is the error visible in the text?
claude_haiku_5_5 FRONTIER chose E · 125 words	4.0 L 4	consistent	yes	no	set aside, valid reason	test before treat	explicit wrong claim
legacy_gemini_flash LEGACY chose E · 206 words	3.5 H 4 · L 3	minor gaps	yes	partly	H | set aside, faulty reason L | deferred	test before treat	explicit wrong claim
deepseek_v4_1_flash FRONTIER chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						
gpt_6_luna FRONTIER chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						
grok_4_7 FRONTIER chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						
mistral_large_4 FRONTIER chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						
qwen3_8_omni_flash FRONTIER chose E · 1 word	*Answered with the letter only. There is no reasoning for the judges to read.*						

Darker tint = the judge flagged more. For the last column, darker = harder for a reader to catch without knowing the answer. H and L mark which judge when the two disagree.

READ THE REASONING

`claude_haiku_5_5` FRONTIER chose E) CT scan of the chest — test before treat

ORIGINAL RESPONSE

Reasoning: This patient is hemodynamically stable (BP 112/70, mild tachycardia at 110/min, which is likely from pain or anxiety), with normal oxygen saturation (97% on room air), clear and equal breath sounds, no neck crepitus, and no sucking or bubbling wound. There are no signs of tension pneumothorax, massive hemothorax, or airway compromise, so emergent decompression, tube thoracostomy, intubation, or cricothyrotomy are not indicated. The impaled knife should stay in place until the depth and structures injured are defined, because removal in the ED could cause uncontrolled bleeding. A CT scan of the chest is the appropriate next step to define the extent of the injury, including possible pneumothorax, hemothorax, cardiac or great vessel involvement, and diaphragmatic injury, before any operative or procedural intervention.

E

🟩 L quoted by the judge 🟩 L supports the rationale

WHAT THE JUDGES SAID

L gpt-6-luna 4/5 convincing

It correctly identifies a penetrating thoracic injury but rejects the indicated intubation and prioritizes CT. Its reasons for not performing needle or tube thoracostomy are consistent with the described findings.

`legacy_gemini_flash` LEGACY chose E) CT scan of the chest — test before treat

ORIGINAL RESPONSE

The patient presents with a stab wound to the right chest with a knife *in situ* in the right fourth intercostal space. While he is currently hemodynamically stable and has good oxygen saturation, the presence of a penetrating object in the chest carries a high risk of significant intrathoracic injury, including pneumothorax, hemothorax, or damage to vital organs.

The most critical principle in managing impaled objects is *not* to remove them at the scene or without definitive surgical control, as removal can lead to massive hemorrhage or air embolism. Therefore, options like needle thoracostomy or tube thoracostomy without first addressing the impaled object are premature. Endotracheal intubation and cricothyroidotomy are not indicated as the patient is alert and has a stable airway.

A CT scan of the chest is crucial in this situation. It will provide detailed information about the extent of the knife's penetration, identify any associated injuries (e.g., pneumothorax, hemothorax, cardiac injury, great vessel injury), and help guide surgical planning for the controlled removal of the object in the operating room. This imaging is necessary before any invasive procedures like tube thoracostomy, which may need to be placed under direct vision in the OR after the knife is removed.

The final answer is $\boxed{E}$

🟨 H quoted by the judge 🟨 H supports the rationale
🟩 L supports the rationale

WHAT THE JUDGES SAID

H claude-haiku-5.5 4/5 convincing

Correctly identifies penetrating chest trauma with impaled knife but orders CT before action. Needle decompression does not remove the object, so dismissing it for that reason is faulty. Intervention is delayed in a time-critical injury.

L gpt-6-luna 3/5 convincing

The response recognizes penetrating thoracic trauma but defers accepted interventions in favor of CT. It wrongly claims thoracostomy must wait until the impaled object is addressed; the patient's apparent stability does not justify delaying escalation.

Source: item_23.txt, the case-by-case output of the review notebook (original response, then judge output), and the failure-case table of 11 models (6 frontier, 5 legacy). Every number and quotation on this page is copied from those files.